PRAXIS Critical Appraisal - When to Stop Giving Fluid (13 Nov 2026)

The clinical question

When septic shock persists after initial resuscitation, is another fluid bolus warranted, or will it add congestion without useful forward flow? This review critically appraises the three trials in the 13 November 2026 PRAXIS Acute Care Physiology newsletter curated by Dr Philippe Rola. It is an independent educational synthesis, not a reproduction of the newsletter.

Key takeaways

- Hypotension is not a diagnosis of hypovolaemia. A MAP target is a safety goal, not proof that tissue perfusion has recovered.
- Fluid responsiveness predicts an increase in stroke volume with a preload challenge. It does not by itself demonstrate ongoing hypoperfusion, clinical need, or fluid tolerance.
- Fluid intolerance is plausible when pulmonary interstitial oedema, high filling pressures, RV dysfunction, or systemic venous congestion coexist with shock.
- Use an iterative perfusion assessment: clinical trajectory, capillary refill time (CRT), lactate trend, bedside echo, dynamic fluid responsiveness testing, lung ultrasound and (when appropriate) venous Doppler. No single POCUS measurement should mandate or prohibit a bolus.
- These three trials do not validate a universal fluid-stopping threshold, a VExUS-directed resuscitation protocol, or a one-size-fits-all vasopressor-first policy.

Trial 1: ARISE FLUIDS - Strategy-level equipoise

Design: Multicentre, open-label randomized ED trial of early vasopressors plus restricted additional crystalloid versus higher fluid volumes with later vasopressors after initial resuscitation. Of 1000 randomized, 963 entered the intention-to-treat population.

Results: At 24 hours, the early-vasopressor strategy used 1108 mL less IV fluid (median difference; 95% CI -1395 to -850). Median days alive and out of hospital at 90 days were 76 in both groups (difference 0 days, 95% CI -2.7 to 2.7; P=1.00). Pulmonary oedema was reported in 0.6% versus 5.0%, respectively. Ninety-day mortality was 16.4% versus 14.4% (RR 1.14, 95% CI 0.85-1.54).

Appraisal: The intervention contrast was credible and the patient-centred primary outcome was neutral. Open-label ascertainment complicates interpretation of pulmonary oedema, and higher ICU admission with early vasopressors is a resource consideration. Trial entry followed initial fluid administration: this was not a test of withholding all early resuscitation fluid. A phenotype-specific benefit is a hypothesis, not an established trial finding.

Trial 2: Melo et al. - Fluid intolerance as a safety limit

Design: Three-centre Chilean ICU randomized trial in mechanically ventilated adults with septic shock, ongoing hypoperfusion, demonstrated fluid responsiveness and at least one risk factor for fluid-related harm. A traffic-light approach integrated lung ultrasound, VExUS, E/lateral e-prime and CVP. Green, yellow and red allowed 500 mL,

250 mL or no bolus, respectively. Recruitment ended early: 36 randomized, 35 analysed (17 intervention, 18 control).

Results: Six-hour protocol bolus volume was 250 mL (IQR 125-500) in the intervention arm versus 650 mL (500-1500) in controls. Total cumulative fluid exposure including pre-randomization fluid was similar: 3375 versus 3750 mL. Change in PaO2/FiO2 at six hours favoured intervention (+40.5 vs -37.9; median between-group difference +78.3, 95% CI 50.5 to 110.0), but the authors explicitly considered the effect uncertain because of premature termination. CRT and lactate changes did not differ.

Appraisal: This tests an attractive integrated safety approach, not whether VExUS alone directs safe fluid resuscitation. With 35 analysed and early stopping, effect estimates are fragile; short-term oxygenation is not mortality or organ recovery. The population was ventilated, preselected for fluid responsiveness and risk of intolerance, and not typical undifferentiated ED shock. Newsletter curator Philippe Rola is a coauthor of the trial; this warrants transparency, not an automatic assumption of bias.

Trial 3: Gao et al. - CRT versus lactate targets

Design: Single-centre, open-label RCT, 85 patients (CRT 44; lactate 41), with six-hour resuscitation after initial treatment. CRT target was <=3 seconds; lactate target was <=2 mmol/L or >10% decrease per hour.

Results: Six-hour total IV input was 1135 mL (IQR 824-1480) in CRT versus 1350 mL (990-1850) in lactate guidance; estimated shift -250 mL (95% CI -500 to -20; P=0.035). No established clinical outcome benefit. The 28-day mortality estimate was imprecise (22.7% vs 29.3%; RR 0.78, 95% CI 0.38-1.60).

Appraisal: Trial was under-recruited for its original mortality sample-size target; the six-hour fluid endpoint had been a secondary outcome and was subsequently prioritized in reporting. Registration occurred retrospectively after enrolment. These details make the apparent fluid difference hypothesis-generating, not a practice-changing finding. It compares multi-intervention target-driven strategies, not CRT as an isolated therapy.

Important external context: ANDROMEDA-SHOCK-2

This larger randomized JAMA trial enrolled 1467 patients and found a favourable hierarchical composite outcome with personalized CRT-targeted haemodynamic resuscitation versus usual care (win ratio 1.16). Its intervention included pulse-pressure and diastolic-pressure assessment, dynamic fluid responsiveness and echo-guided choices of fluid, vasopressors or inotropes. This supports structured, physiology-informed reassessment more convincingly than the small Gao trial, but does not establish that every single CRT, VExUS or 'four-interface' construct is independently validated.

Practical ED/resuscitation sequence

1. Treat immediately reversible threats: oxygenation/ventilation, source control and antimicrobials, life-threatening hypotension, haemorrhage, arrhythmia and obstructive pathology.

2. Assess whether hypoperfusion persists: mentation, skin temperature and CRT, urine-output trajectory, serial lactate in context, and overall clinical trend. Consider other causes of elevated lactate.

3. Identify likely limitation: low preload, vasoplegia, LV dysfunction, RV/pulmonary vascular constraint, or excess systemic venous pressure. Interpret cardiac and lung POCUS together.

4. If considering fluid, require a coherent reason: evidence of inadequate perfusion plus plausible preload dependence. Prefer dynamic assessment of stroke volume response when feasible.

5. Check tolerance before and after any cautious fluid challenge: worsening oxygenation/B-lines, rising venous congestion, severe RV dysfunction or increased filling pressures should prompt reassessment of benefit versus harm.

6. If fluid is unlikely to help or poorly tolerated, prioritize mechanism-specific alternatives such as norepinephrine for vasoplegia and targeted treatment of cardiac or obstructive causes. Reassess perfusion after every intervention.

Bottom line

The PRAXIS message 'treat the physiology, not the number' is clinically useful. But bedside physiology is a decision framework, not a licence to substitute attractive surrogates for validated outcomes. ARISE FLUIDS is neutral on strategy superiority; the fluid-intolerance trial is promising but underpowered; the Gao trial is exploratory. Use these findings to become more selective and iterative with fluids, not to abolish fluids or apply an unvalidated congestion cutoff.

Primary references

1. ARISE FLUIDS Investigators. Vasopressors or Fluids in Early Septic Shock. N Engl J Med. 2026;395:1387-1398. doi:10.1056/NEJMoa2516225.

2. Melo P, Amthauer M, Ferri G, et al. Fluid-intolerance signals as bedside safety limits during early septic shock resuscitation: a multicenter randomized trial. Crit Care. 2026;30:574. doi:10.1186/s13054-026-06316-y.

3. Gao F, Xu Y, Li Z, et al. Effect of capillary refill time-guided resuscitation on early fluid administration and fluid balance in septic shock: a single-center randomized controlled trial. Sci Rep. Published 26 October 2026. doi:10.1038/s41598-026-77226-w.

4. ANDROMEDA-SHOCK-2 Investigators. Personalized Hemodynamic Resuscitation Targeting Capillary Refill Time in Early Septic Shock. JAMA. 2025;334:1988-1999. doi:10.1001/jama.2025.20402.

Source and evidence level

Newsletter: PRAXIS Acute Care Physiology, issue 2 (13 November 2026), 'When to Stop Giving Fluid.' Appraisal based on newsletter text and independently cross-checked against published primary article abstracts and accessible full-text sections. The trial evidence is distinguished from physiological interpretation and bedside extrapolation.